

Question 1: Heuristic design and admissibility

Steps:

Heuristic is the number of positions in the current word that are different from the goal word. Each action can only change one letter. Therefore, if there are three letters that do not match the goal, at least three actions are required.

It cannot overstimulate the remaining cost because every mismatched position will be corrected, and every step costs 1.

Scrabble:

For every position that does not match the goal, I add the Scrabble value of the goal letter for that position. For example, c=3; a=1; t=1, so the heuristic value is 5.

I used this because every incorrect position has eventually to be changed into the correct goal letter. At some point the Scrabble cost of inserting that letters need to be paid.

Frequency

For frequency I used $h(n) = \text{number of mismatched letters} + \text{rarity (goal)}$

If the current word is already the goal, then the heuristic is 0. The reason I used this is because every mismatched position needs at least one move to fix it. Every frequency move costs at least 1 because the formula is: **1 + rarity(word)**

Because of that, I am only counting costs that are guaranteed to happen, so the heuristic should not overestimate the real cost.

Question 2: Does the Heuristic Actually Help?

1. Yes, in my tests UCS and A* returned the same optimal cost for all the problems. This makes sense because my heuristics are admissible, so A* still finds the optimal solution.
2. For both WHY -> ASK and BEAR -> DUCK, Scrabble had the largest percentage reduction in expanded nodes in my tests.
3. The frequency heuristic was the least informative for both word pairs.
4. I think steps worked well because it basically tells us how many letters, we still need to change to reach the goal. Scrabble gives us a little more information because it also looks at how much each of those letters will cost to change.

Word Pair	Cost	UCS Cost	UCS Expanded	A* Cost	A* Expanded	Reduction
WHY → ASK	Steps	5.00	773	5.00	16	97.93%
WHY → ASK	Scrabble	9.00	765	9.00	13	98.30%
WHY → ASK	Frequency	38.86	808	38.86	634	21.53%
BEAR → DUCK	Steps	4.00	691	4.00	6	99.13%
BEAR → DUCK	Scrabble	11.00	2981	11.00	6	99.80%
BEAR → DUCK	Frequency	38.24	1084	38.24	302	72.14%

Question 3: Reversibility and Direction Dependent Costs

1. Steps

Yes, the cost should be the same going in either direction. Every move costs 1, so if it takes 5 moves to go from one word to another, reversing the same path would also take 5 moves. Therefore, the optimal steps cost should be the same in both directions.

2. Scrabble

No, the action can be reversed, but that does not mean the cost is the same.

3. Frequency

No, the frequency is also not necessarily the same in both directions because the cost depends on the word we are entering. Therefore, the actions can be reversible without the costs being equal in both directions.

Questions 4: LLM Heuristic Audit

1. I used ChatGPT with GPT 5.6 Sol. $h(n)$ = number of mismatched letters + rarity(goal)

2. If state == goal:

return 0

different = number of letters that do not match the goal

return different + rarity(goal)

3. Yes.

If there are d mismatched letters, then we need at least d actions because one action can only change one letter.

Every frequency action costs:

$$1 + \text{rarity}(\text{word})$$

Since rarity is nonnegative, every move costs at least 1.

Also, any successful solution eventually has to enter the goal word, so we know we have to pay the rarity of the goal at least once.

Because of this, $d + \text{rarity}(\text{goal})$ should always be less than or equal to the real remaining cost, so it is admissible.

4. No.

I do not think the heuristic needs to be fixed because it is already admissible.

It is also more informative than the original heuristic:

$$h_0(n) = \text{number of mismatched letters}$$

because it adds the rarity value of the goal on top of that.

5. The original heuristic expanded **755 nodes**.

The rarity-aware heuristic expanded **634 nodes**.

Both returned the same optimal cost of **38.86**.

For BEAR $\rightarrow$ DUCK:

The original heuristic expanded **720 nodes**.

The rarity-aware heuristic expanded **302 nodes**.

Both returned the same optimal cost of **38.24**.

Based on these tests, the rarity-aware heuristic worked better because it expanded fewer nodes while still returning the same optimal cost.

However, I do not think the tests alone prove that the heuristic is admissible. The proof comes from the fact that it is only counting unavoidable costs.

I used ChatGPT with GPT-5.6 Sol to help me understand parts of the assignment, debug my code, check my heuristic reasoning, and organize my answers. I verified the code using the provided testing files and compared the results with the examples from the assignment.